

## ONslow, Australia

WMO: 943050

Lat: 21.633S

Lon: 115.117E

Elev: 13

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-04

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 7             | 53.2       | 55.1 | 32.9                          | 27.4 | 66.3 | 36.7 | 31.9 | 67.9 | 20.6                  | 66.1 | 18.1 | 68.8 | 4.1                      | 180  | 0.367 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 13.9                   | 101.7           | 72.2 | 97.7 | 74.3 | 94.5 | 75.8 | 83.4                | 90.5 | 81.9 | 89.9 | 80.9 | 88.9 | 10.0                 | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.7                            | 164.7 | 88.7 | 79.7 | 154.4 | 87.5 | 78.7 | 149.1 | 86.6 | 47.5          | 90.6 | 45.7 | 90.0 | 44.7 | 89.1 | 97.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 20.4              | 18.4 | 16.5 | DB                         | 48.6 | 110.5              | 1.7 | 3.6                                                | 47.4 | 113.0      | 46.4 | 115.1      | 45.4 | 117.1      | 44.2 | 119.7 |  |
|                   |      |      | WB                         | 43.1 | 88.4               | 2.1 | 5.2                                                | 41.6 | 92.1       | 40.4 | 95.2       | 39.2 | 98.1       | 37.7 | 101.8 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 76.6   | 84.3  | 85.2  | 84.7  | 80.6  | 74.6  | 68.3  | 66.7  | 68.3  | 70.7  | 75.7  | 78.0  | 82.1  |
|                                                                        | DBStd       | 7.55   | 4.27  | 3.43  | 3.24  | 3.07  | 3.75  | 3.64  | 3.40  | 3.00  | 3.08  | 4.60  | 4.02  | 4.39  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 38     | 0     | 0     | 0     | 0     | 0     | 9     | 21    | 7     | 1     | 0     | 0     | 0     |
|                                                                        | CDD50       | 9693   | 1064  | 986   | 1076  | 917   | 762   | 548   | 519   | 568   | 621   | 798   | 840   | 994   |
|                                                                        | CDD65       | 4256   | 599   | 566   | 611   | 467   | 297   | 107   | 75    | 110   | 172   | 333   | 390   | 529   |
|                                                                        | CDH74       | 45665  | 7783  | 7508  | 7906  | 4945  | 2146  | 440   | 280   | 585   | 1095  | 2976  | 3749  | 6252  |
|                                                                        | CDH80       | 19574  | 3817  | 3769  | 3968  | 1939  | 514   | 28    | 24    | 64    | 198   | 1088  | 1378  | 2787  |
| (14) Wind                                                              | WSAvg       | 7.7    | 9.5   | 8.8   | 7.8   | 5.8   | 5.3   | 5.3   | 5.7   | 6.3   | 8.2   | 9.3   | 10.3  | 10.7  |
| (15) Precipitation                                                     | PrecAvg     | 11.3   | 1.0   | 2.9   | 2.0   | 0.7   | 1.8   | 1.9   | 0.5   | 0.3   | 0.1   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecMax     | 39.3   | 5.7   | 21.5  | 14.5  | 4.6   | 9.0   | 8.9   | 3.5   | 2.4   | 1.0   | 1.1   | 0.9   | 0.7   |
|                                                                        | PrecMin     | 3.0    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 7.9    | 1.5   | 5.2   | 3.2   | 1.1   | 2.1   | 2.3   | 0.7   | 0.6   | 0.2   | 0.2   | 0.2   | 0.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 108.2 | 105.0 | 102.6 | 95.9  | 89.8  | 83.0  | 83.0  | 85.2  | 90.1  | 101.5 | 102.2 | 105.9 |
|                                                                        |             | MCWB   | 71.2  | 73.4  | 74.1  | 74.0  | 72.0  | 67.7  | 64.1  | 63.5  | 64.5  | 67.6  | 67.6  | 71.1  |
|                                                                        | 2%          | DB     | 99.8  | 98.2  | 97.7  | 92.4  | 87.1  | 80.5  | 79.3  | 81.9  | 85.3  | 93.8  | 94.5  | 98.0  |
|                                                                        |             | MCWB   | 73.8  | 77.0  | 76.7  | 74.9  | 71.8  | 67.3  | 63.7  | 63.9  | 64.0  | 67.2  | 68.4  | 72.4  |
|                                                                        | 5%          | DB     | 95.6  | 94.1  | 94.7  | 90.3  | 85.0  | 78.6  | 76.9  | 79.5  | 82.0  | 88.7  | 89.7  | 92.8  |
|                                                                        |             | MCWB   | 75.8  | 79.3  | 78.0  | 74.6  | 71.4  | 66.5  | 64.3  | 63.9  | 65.3  | 67.2  | 69.9  | 73.9  |
|                                                                        | 10%         | DB     | 92.0  | 91.8  | 92.2  | 88.2  | 83.0  | 76.8  | 75.2  | 77.5  | 79.8  | 85.3  | 86.8  | 89.9  |
|                                                                        |             | MCWB   | 77.5  | 79.5  | 78.6  | 74.5  | 70.9  | 65.6  | 64.0  | 64.0  | 65.9  | 67.7  | 71.2  | 76.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 82.9  | 85.3  | 91.6  | 81.4  | 78.7  | 74.3  | 72.5  | 71.1  | 72.5  | 76.2  | 77.3  | 81.9  |
|                                                                        |             | MCDB   | 91.1  | 93.3  | 93.1  | 86.9  | 84.2  | 78.7  | 75.5  | 76.6  | 78.6  | 84.1  | 86.3  | 89.6  |
|                                                                        | 2%          | WB     | 81.6  | 82.8  | 85.1  | 79.0  | 76.6  | 72.3  | 69.7  | 69.3  | 71.0  | 74.1  | 76.1  | 80.1  |
|                                                                        |             | MCDB   | 90.2  | 91.0  | 88.5  | 86.1  | 82.1  | 76.8  | 74.2  | 75.5  | 77.8  | 82.3  | 85.1  | 88.4  |
|                                                                        | 5%          | WB     | 80.6  | 81.8  | 82.1  | 78.0  | 75.2  | 70.6  | 67.9  | 67.9  | 69.8  | 72.6  | 75.0  | 79.0  |
|                                                                        |             | MCDB   | 88.9  | 89.9  | 89.8  | 85.5  | 80.7  | 75.1  | 73.4  | 74.2  | 77.0  | 81.5  | 83.9  | 87.5  |
|                                                                        | 10%         | WB     | 79.7  | 81.0  | 80.8  | 77.0  | 73.8  | 68.6  | 66.2  | 66.4  | 68.4  | 71.1  | 73.9  | 78.0  |
|                                                                        |             | MCDB   | 88.0  | 89.0  | 88.5  | 84.9  | 80.1  | 73.5  | 71.9  | 73.6  | 76.1  | 80.7  | 82.5  | 86.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.2  | 13.9  | 16.6  | 16.7  | 16.9  | 17.1  | 18.3  | 20.0  | 20.4  | 21.0  | 18.3  | 16.1  |
|                                                                        |             | MCDBR  | 23.8  | 19.2  | 21.2  | 19.1  | 18.8  | 18.6  | 20.2  | 22.6  | 23.7  | 26.9  | 24.9  | 22.8  |
|                                                                        |             | MCWBR  | 12.0  | 9.1   | 10.6  | 10.4  | 10.4  | 10.1  | 11.0  | 12.1  | 12.2  | 12.3  | 12.2  | 11.2  |
|                                                                        | 5% WB       | MCDBR  | 14.7  | 13.4  | 15.6  | 15.3  | 14.4  | 13.1  | 15.8  | 17.3  | 18.5  | 19.5  | 17.3  | 14.6  |
|                                                                        |             | MCWBR  | 9.6   | 7.5   | 10.3  | 9.6   | 9.0   | 7.9   | 10.2  | 11.2  | 11.5  | 12.7  | 10.4  | 9.8   |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.394  | 0.395 | 0.375 | 0.352 | 0.331 | 0.322 | 0.314 | 0.317 | 0.329 | 0.346 | 0.351 | 0.367 | (40)  |
|                                                                        | taud        | 2.513  | 2.527 | 2.573 | 2.597 | 2.599 | 2.608 | 2.607 | 2.561 | 2.521 | 2.496 | 2.512 | 2.533 | (41)  |
|                                                                        | Ebn at Noon | 302    | 297   | 295   | 290   | 284   | 281   | 287   | 297   | 305   | 309   | 313   | 310   | (42)  |
|                                                                        | Edh at Noon | 36     | 35    | 32    | 29    | 27    | 25    | 26    | 30    | 33    | 36    | 36    | 36    | (43)  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 2447   | 2315  | 2041  | 1722  | 1438  | 1289  | 1390  | 1734  | 2080  | 2373  | 2561  | 2591  | (44)  |
|                                                                        | RadStd      | 118    | 129   | 128   | 97    | 105   | 90    | 103   | 57    | 68    | 65    | 70    | 107   | (45)  |

## Historical Trends

|                        | DBAvg      | Heating    |            | Cooling    |            |            | Degree-Days |            |            |            |
|------------------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|------------|
|                        |            | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD50       | HDD65      | CDD50      | CDD65      |
| Station Only           | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A  | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A |
| Regional (6 neighbors) | +0.56      | N/A        | -1.84      | N/A        | N/A        | N/A        | N/A         | N/A        | +275       | +288       |

Nomenclature: See separate page